

Yonghoon Jeong

Lecturer (Assistant Professor-track), Department of Cyber Science
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Research Interests

Wireless Communication Systems
Physical-Layer Wireless Sensing and Networking
Low-Power Wide-Area Networks (LPWANs)
Underwater Acoustic Communications
Ocean IoT

Education

M.S. in Electrical and Computer Engineering Seoul National University, Seoul, South Korea Advisor: Prof. Saewoong Bahk	2024 – 2026
B.S. in Electronics and Communications Engineering Kwangwoon University, Seoul, South Korea	2018 – 2023

Professional Experience

Republic of Korea Naval Academy (ROKNA) , Changwon, South Korea Lecturer (Assistant Professor-track), Department of Cyber Science Researcher, Institute of Naval Research	2026 – Present
Seoul National University (SNU) , Seoul, South Korea Researcher, Network Lab (NETLAB), Institute of New Media and Communications (INMC)	2024 – 2026
Electronics and Telecommunications Research Institute (ETRI) , Pangyo, Seongnam, South Korea Researcher, AI-SoC Infrastructure Research Section	2023
Korea Advanced Institute of Science and Technology (KAIST) , Daejeon, South Korea Research Intern, Unmanned Systems Research Group (USRG), School of Electrical Engineering (EE) Advisor: Prof. David Hyunchul Shim	2022

Publications

International Papers

“Receiver-Aware Chirp Waveform Optimization for Mobile Underwater LoRa Communications,” in preparation.
“On the Symbol Error Rate of Non-Linear Chirp Modulation for LoRa Under Asynchronous Interference,” under review, *IEEE Wireless Communications Letters*. (SCIE, IF: 5.5)
Yonghoon Jeong, Youngwook Son, and Saewoong Bahk, “On the Interference Resilience of Non-Linear Chirp Modulation for LoRa: An Analytical Insight into Energy Scattering Effect,” *IEEE Communications Letters*, vol. 30, pp. 2238–2242, Jun. 2026. (SCIE, IF: 4.5) [Link](#)
Dongrakk Choi, Yubin Choi, **Yonghoon Jeong**, Jeongyeup Paek, and Saewoong Bahk, “D²-PLC: Holistic Design and Implementation of High-Data-Rate Duplex DC Power Line Communication Network,” *IEEE Internet of Things Journal*, vol. 12, pp. 45325–45340, Nov. 2025. (SCIE, IF: 8.9) [Link](#)
Yonghoon Jeong and Youngseok Choi, “Diffusion-Phys: Noise-Robust Heart Rate Estimation from Facial Videos via Diffusion Models,” *Biomedical Engineering Letters*, vol. 15, pp. 575–585, Apr. 2025. (SCIE, IF: 3.4) [Link](#)
Dongrak Choi, **Yonghoon Jeong**, Yubin Choi, and Saewoong Bahk, “A Programmable High-Throughput Duplex

DC-PLC Testbed for Power and Data Integration,” *IEEE ICNP 2025*, Seoul, Korea, Oct. 19–22, 2025. [Link](#)

Domestic Papers

Yonghoon Jeong and Saewoong Bahk, “A Study on Direct Current Power Line Communication (DC-PLC) with Applications in Automotive, Maritime, and Aviation,” *The Conference of the Korea Institute of Communications and Information Sciences (KICS)*, Gyeongju, Korea, Nov. 20–22, 2024.

Yonghoon Jeong, “Development of Non-contact Heart Rate Measurement Technology Based on Facial Video,” *The Conference of the Korean Institute of Information Scientists and Engineers (KIISE)*, Jeju, Korea, Jun. 23–25, 2021.

Awards & Scholarships

Scholarship Recipient	2024–2025
Brain Korea 21 (BK21), National Research Foundation of Korea	
Scholarship Recipient	2022–2023
Korea Scholarship Foundation for the Future Leaders (KOSFFL)	
Excellent Paper Awards	2021
Korea Computer Conference 2021	
University President’s Award	2020
Seoul Capstone Design, Seoul	
Scholarship Recipient	2020
Samsung Scholarship Foundation	
Dean’s List	2020
Highest GPA	

Teaching Experience

Creative Engineering Design , Prof. Saewoong Bahk	2024
Electronic Circuits Experiment I , Emeritus Prof. Young-chul Chung	2023
Electronic Circuits Experiment II , Emeritus Prof. Young-chul Chung	2023
Electronic Circuits Experiment I , Prof. Youngseok Choi	2022

Technical Skills

Programming Languages: C, C++, Python, MATLAB

Frameworks & Libraries: ROS, ROS2, TensorFlow, PyTorch

Simulation Tools: MATLAB/Simulink, ns-3

Development Tools: Linux, Git, Docker